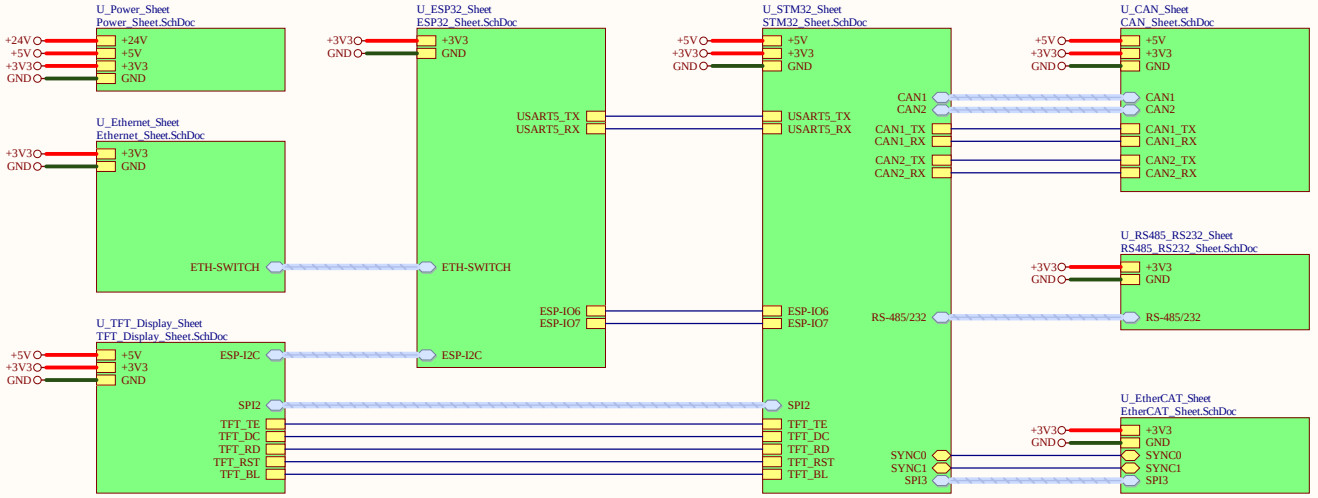
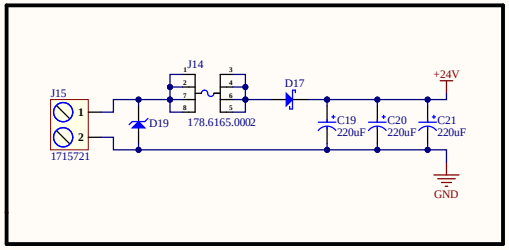
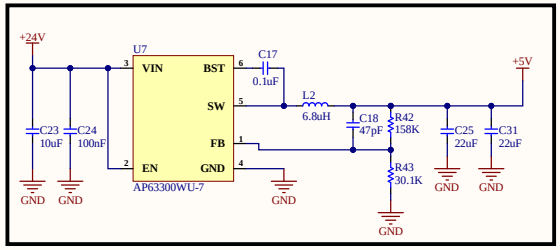


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Date: 16-Oct-24	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 9

24V INPUT

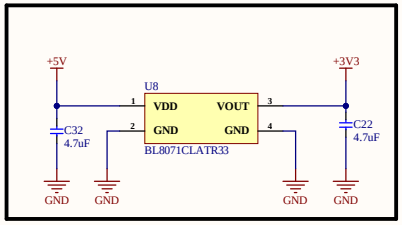


5V 3A BUCK



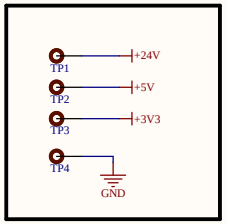
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3.3V 1.5A BUCK

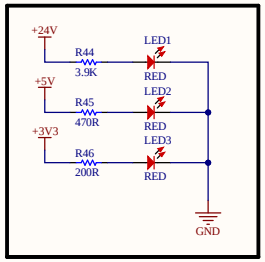


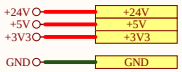
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TEST POINTS



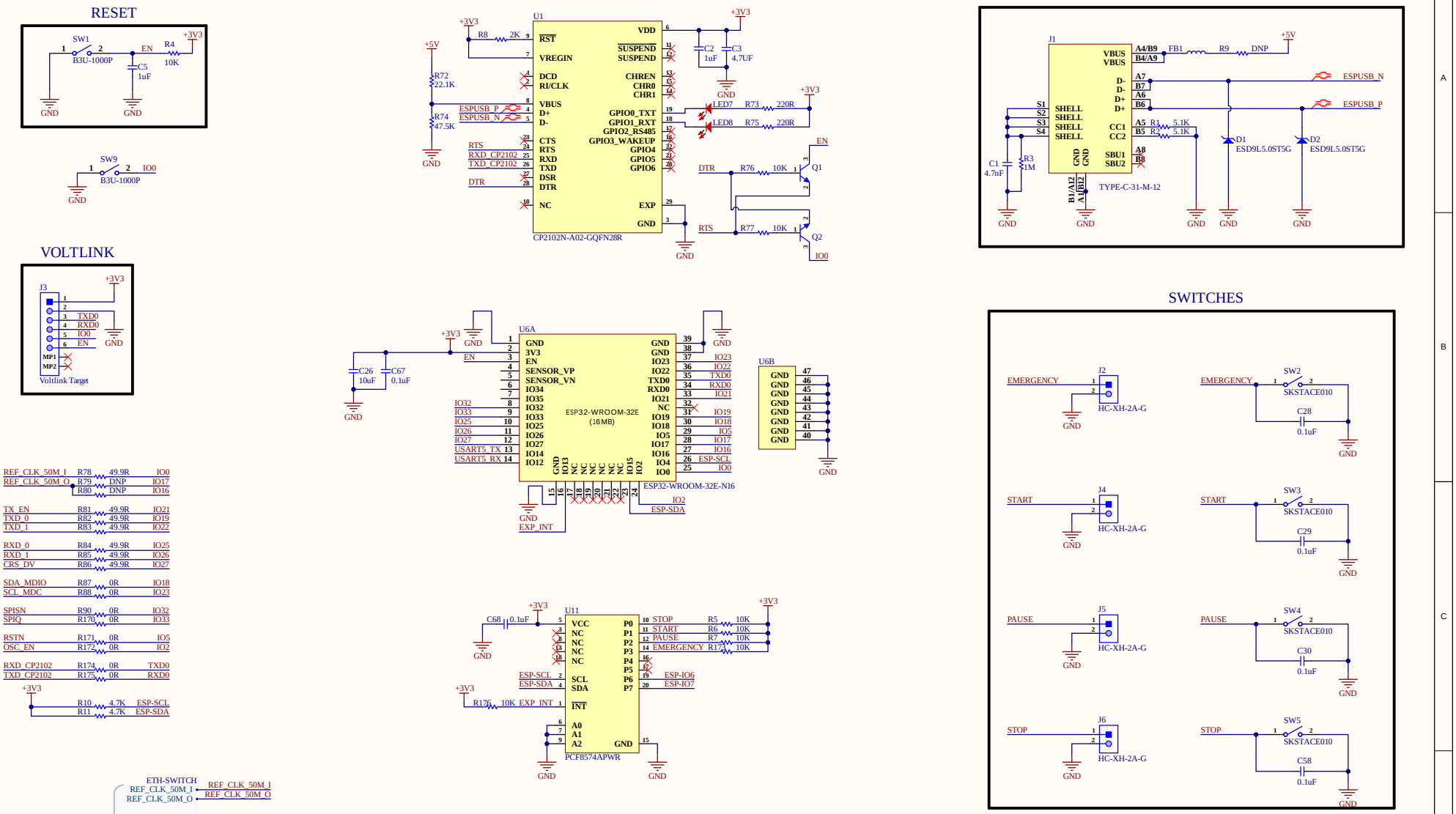
LEDs



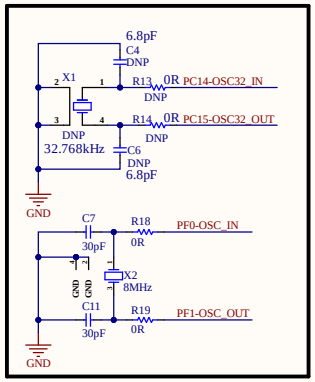


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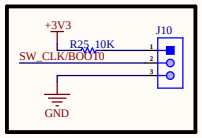
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File: Control-Board-v1.0.PrjPcb	
Date: 27-Sep-24	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 9


RESET
SW1
B3U-1000P
EN
R4
10K
+3V3
C5
1uF
GND
SW9
IO0
VOLTLINK
J3
TXD0
RXD0
EN
MP1
MP2
VoltLink Target
U1
CP2102N-A02-GQFN28R
R8
2K
R72
22.1K
R74
47.5K
ESPUSB_P
ESPUSB_N
RTS
RXD_CP2102
TXD_CP2102
DTR
LED7
LED8
R73
R75
220R
R76
R77
10K
Q1
Q2
C2
1uF
C3
4.7UF
U6A
ESP32-WROOM-32E
(16MB)
ESP32-WROOM-32E-N16
U6B
C26
10uF
C67
0.1uF
IO32
IO33
IO25
IO26
IO27
USART5_TX
USART5_RX
EXP_INT
ESP-SDA
REF_CLK_50M_I
REF_CLK_50M_O
R78
R79
R80
DNP
TX_EN
TXD_0
TXD_1
RXD_0
RXD_1
CRS_DV
SDA_MDIO
SCL_MDC
SPISN
SPIQ
RSTN
OSC_EN
R81
R82
R83
R84
R85
R86
R87
R88
R90
R170
R171
R172
R175
R10
R11
4.7K
ESP-SCL
U11
PCF8574APWR
C68
0.1uF
R176
EXP_INT
ESP-IO6
ESP-IO7
STOP
START
PAUSE
EMERGENCY
R5
R6
R7
R174
ETH-SWITCH
USB
J1
TYPE-C-31-M-12
FB1
R9
DNP
R1
R2
5.1K
C1
4.7nF
R3
1M
D1
D2
ESD9L5.0ST5G
ESPUSB_N
ESPUSB_P
SWITCHES
J2
J4
J5
J6
HC-XH-2A-G
SW2
SW3
SW4
SW5
SKSTACE010
C28
C29
C30
C58
0.1uF
ESP-I2C
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Date: 17-Oct-24
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Drawn By: Huzaifa Ali
Page Total: 9

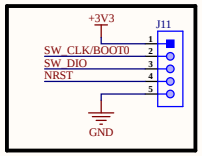
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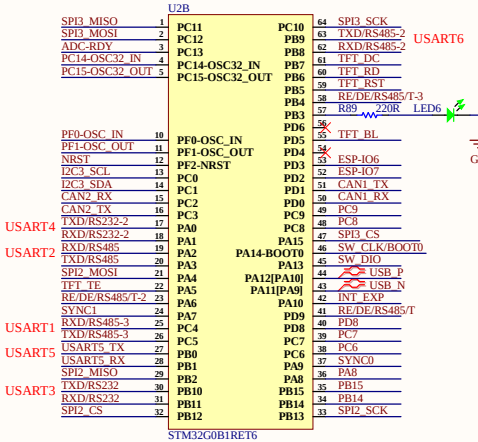


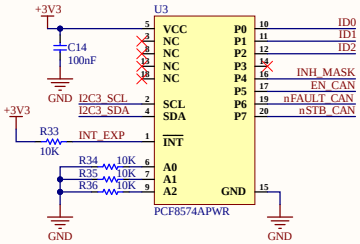
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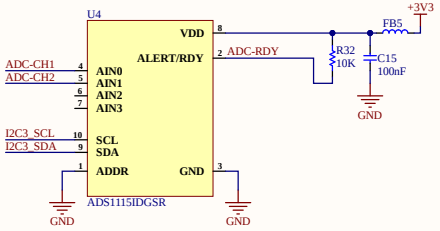


SWD PORT

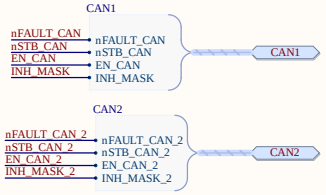


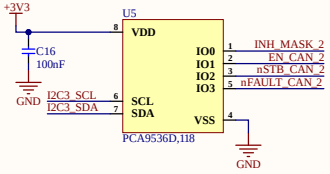


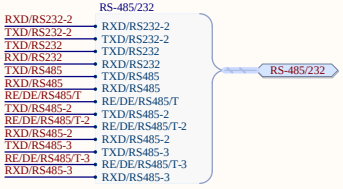


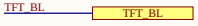


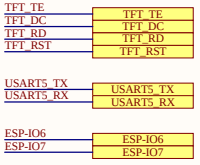


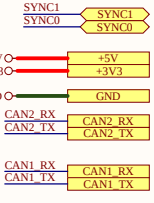




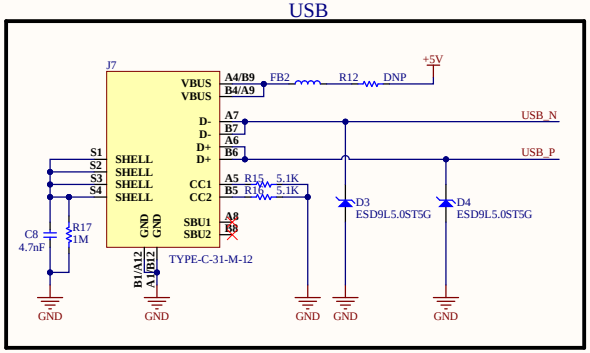


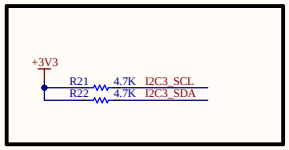




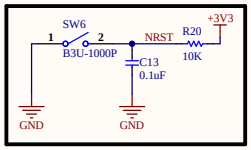


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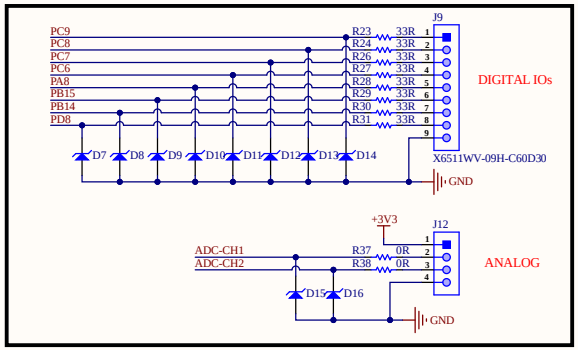


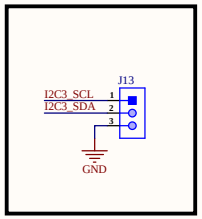


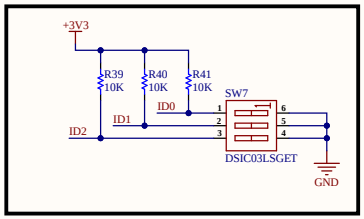
RESET



DIO ADC



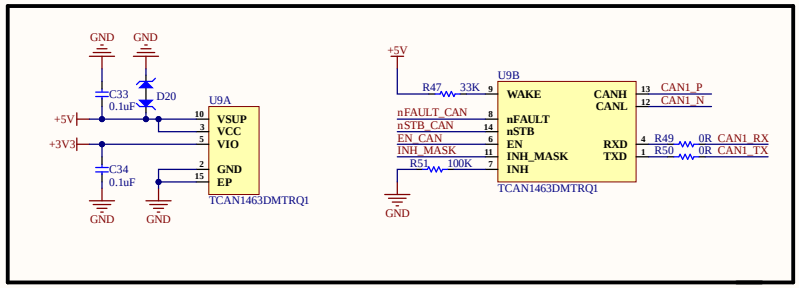




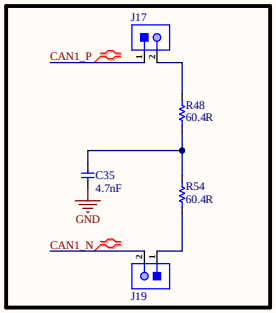
eSpark Consulting Group

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Date: 27-Sep-24	Page	Num: 4
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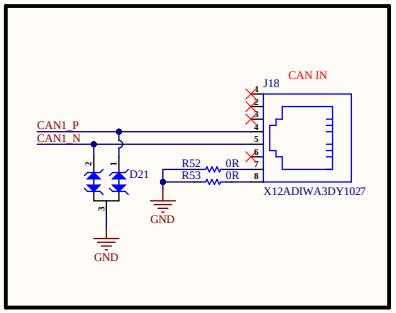
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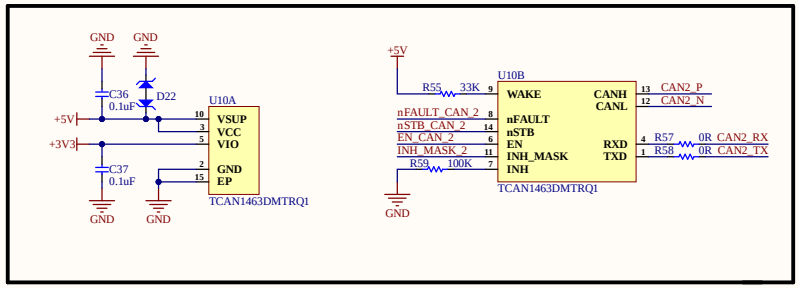
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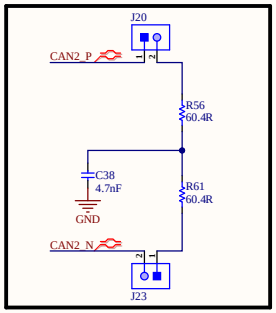
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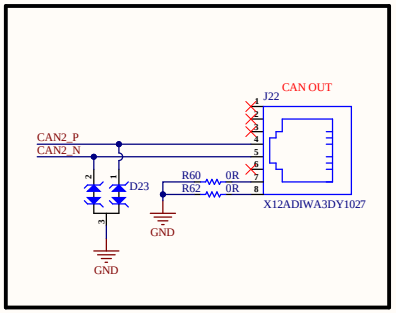
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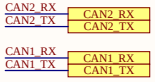


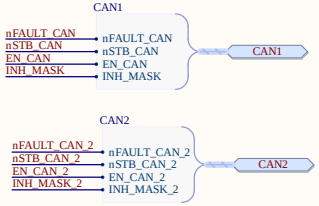
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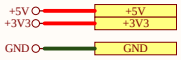


CAN2 CONNECTOR





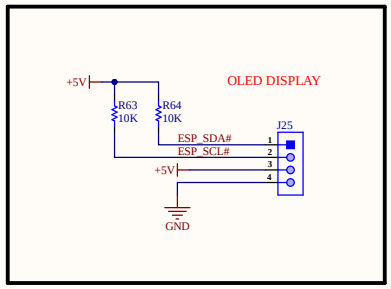




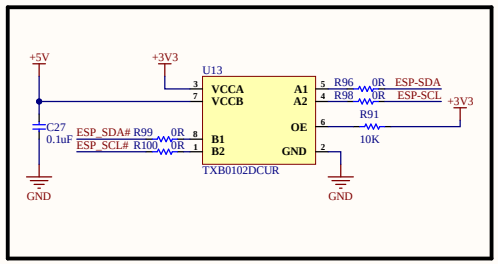
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Date: 09-Sep-24	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 9

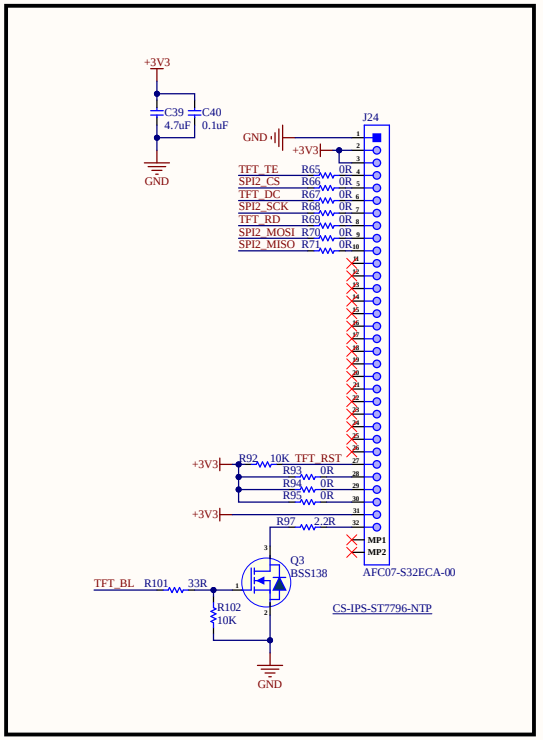
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LEVEL SHIFTER

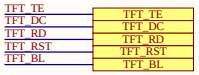


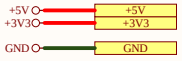
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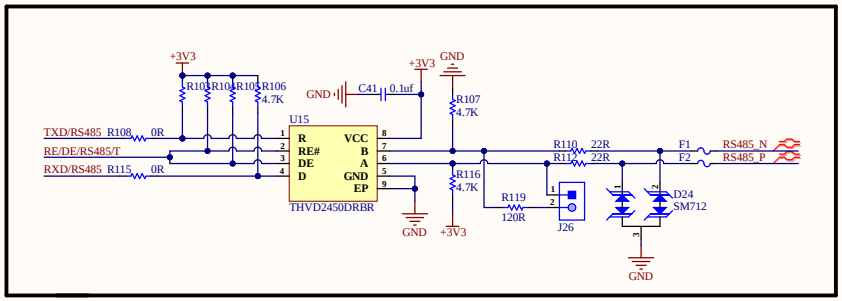




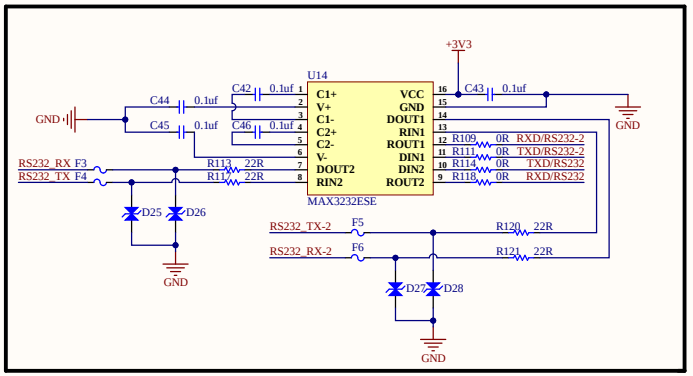
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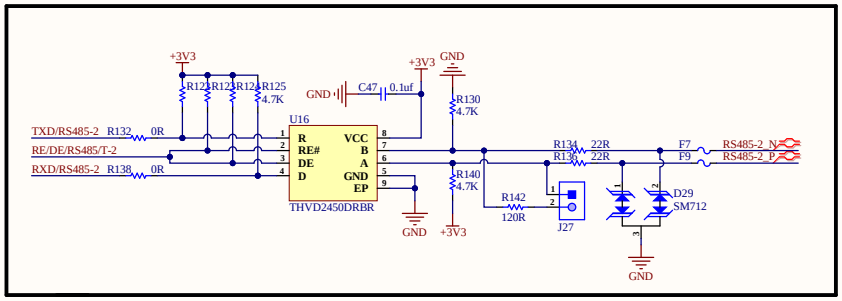
RS485 TO UART



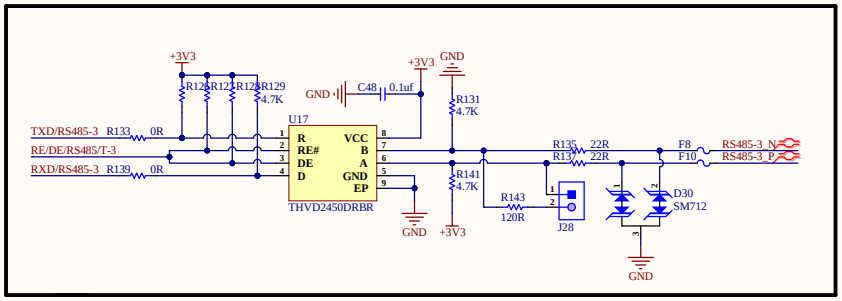
RS232 TO UART

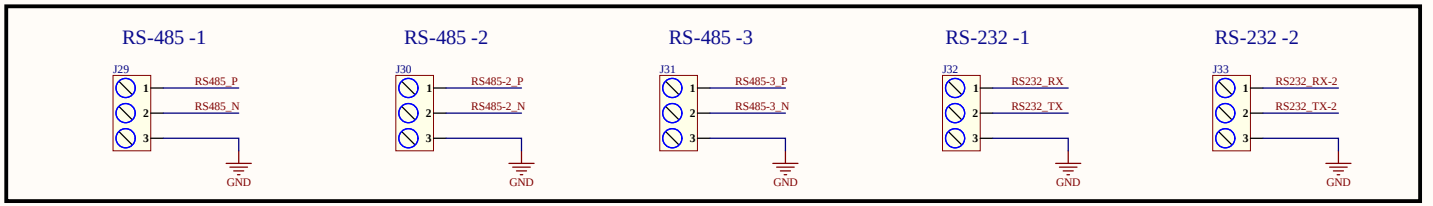


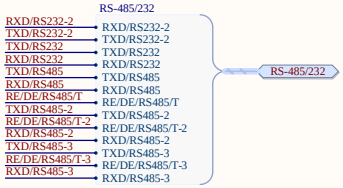
RS485 TO UART



RS485 TO UART



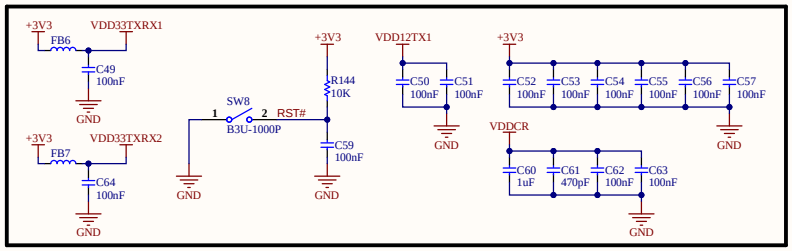




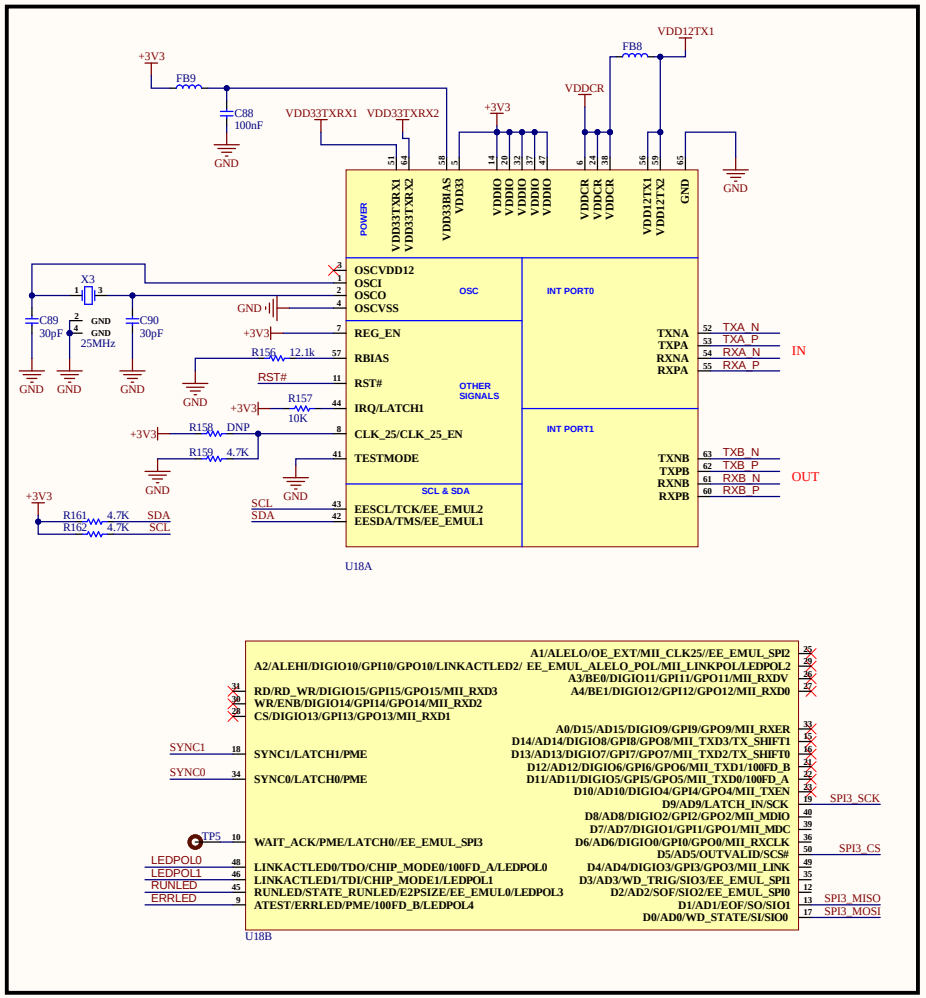


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Date: 27-Sep-24	Page Num: 7
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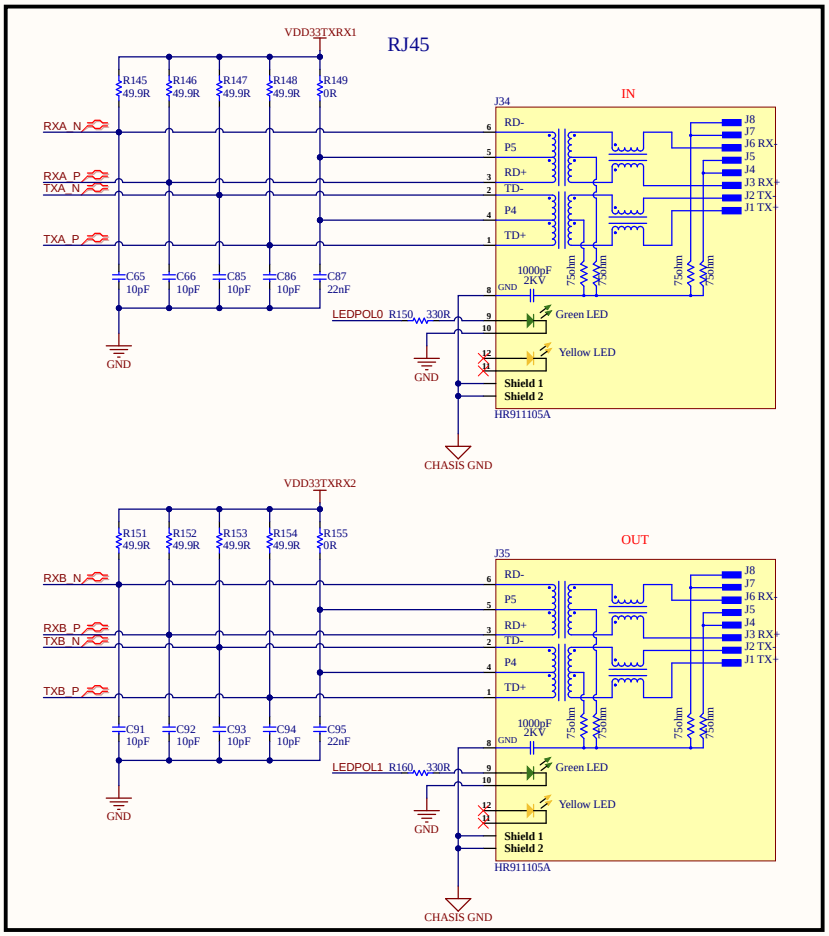


ETHERCAT

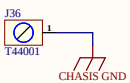


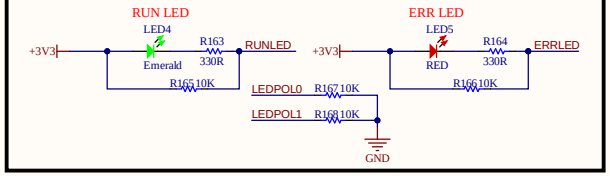


RJ45

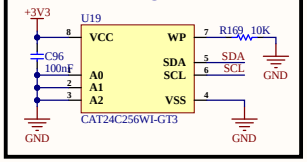


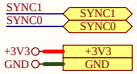
EARTH TERMINAL





EEPROM

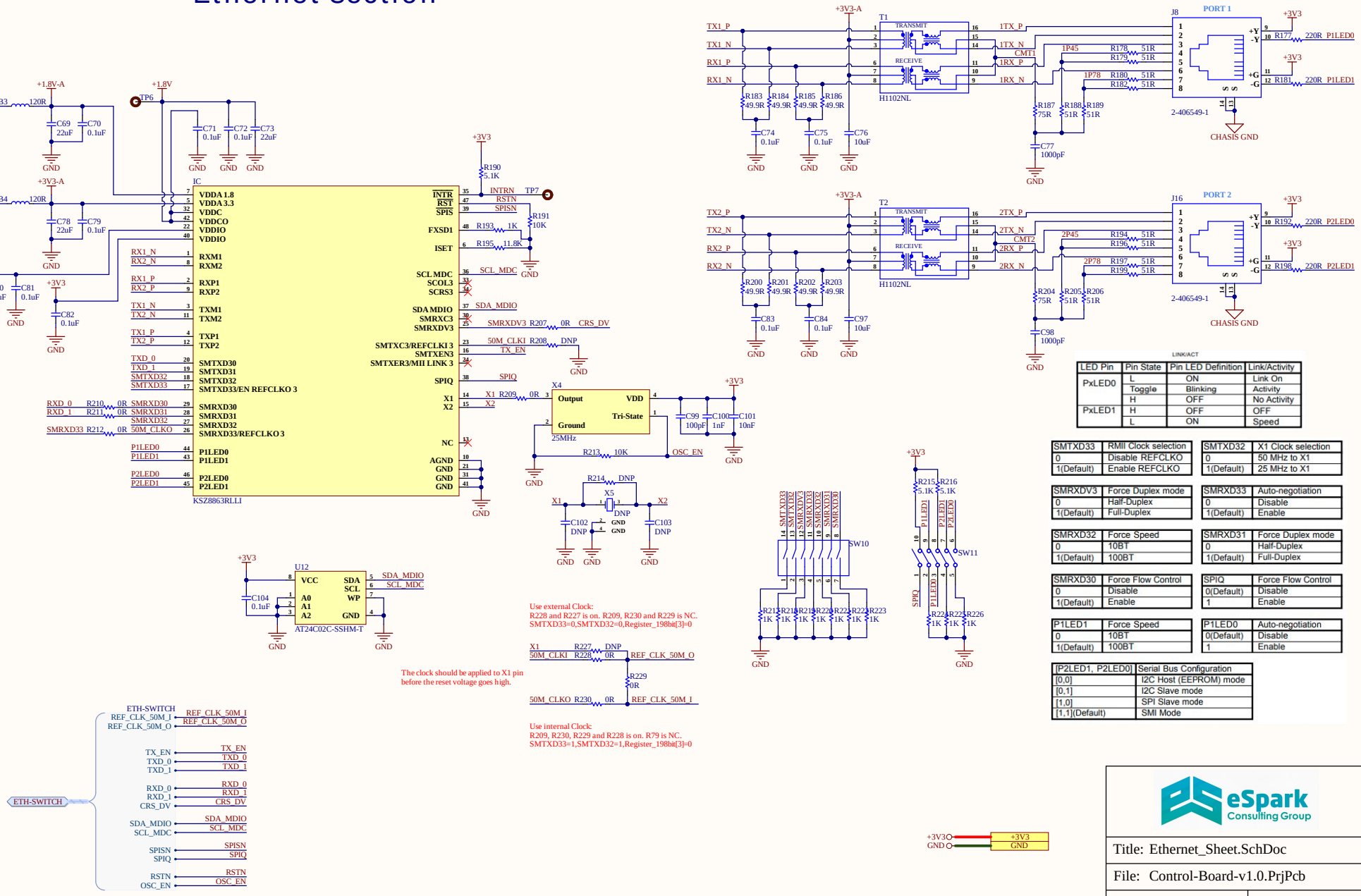






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Ethernet Section



Use external Clock:
R228 and R227 is on. R209, R230 and R229 is NC.
SMTXD33=0,SMTXD32=0,Register_198bit[3]=0

The clock should be applied to X1 pin before the reset voltage goes high.

Use internal Clock:
R209, R230, R229 and R228 is on. R79 is NC.
SMTXD33=1,SMTXD32=1,Register_198bit[3]=0

LINK/ACT

LED Pin	Pin State	Pin LED Definition	Link/Activity
PxLED0	L	ON	Link On
	Toggle	Blinking	Activity
	H	OFF	No Activity
PxLED1	H	OFF	OFF
	L	ON	Speed

SMTXD33	RMII Clock selection
0	Disable REFCLKO
1(Default)	Enable REFCLKO

SMTXD32	X1 Clock selection
0	50 MHz to X1
1(Default)	25 MHz to X1

SMRXDV3	Force Duplex mode
0	Half-Duplex
1(Default)	Full-Duplex

SMRXD33	Auto-negotiation
0	Disable
1(Default)	Enable

SMRXD32	Force Speed
0	10BT
1(Default)	100BT

SMRXD31	Force Duplex mode
0	Half-Duplex
1(Default)	Full-Duplex

SMRXD30	Force Flow Control
0	Disable
1(Default)	Enable

SPIQ	Force Flow Control
0(Default)	Disable
1	Enable

P1LED1	Force Speed
0	10BT
1(Default)	100BT

P1LED0	Auto-negotiation
0(Default)	Disable
1	Enable

[P2LED1, P2LED0]	Serial Bus Configuration
[0,0]	I2C Host (EEPROM) mode
[0,1]	I2C Slave mode
[1,0]	SPI Slave mode
[1,1](Default)	SMI Mode

eSpark Consulting Group

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